Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 5: Approved Solutions

Converse, Contrapositive, and the Language of "If"

Covers Chartrand & Zhang, Section 1.3, pp. 28–32

Status of this set

Ungraded practice. Solutions released with it. Attempt everything before reading any solution.

Part I: How to Practice This Material

Two questions to ask every time

The drill

When you meet an implication, ask:

1. **Which way does the arrow point?** English hides this. "P only if Q" and "Q if P" both mean $P \Rightarrow Q$, even though Q is written first in one of them.
2. **Am I being asked about the statement or its converse?** They are different claims, and confusing them is the most consequential error in Chapter 1.

The pairing, in one line

$$P \Rightarrow Q \;\equiv\; (\sim Q) \Rightarrow (\sim P) \qquad \text{and} \qquad Q \Rightarrow P \;\equiv\; (\sim P) \Rightarrow (\sim Q)$$

Implication pairs with contrapositive; converse pairs with inverse. A single counterexample kills both members of a pair at once.

Three mistakes to expect

1. **Reading "P only if Q" as $Q \Rightarrow P$.** It is $P \Rightarrow Q$.
2. **Swapping "necessary" and "sufficient."** In $P \Rightarrow Q$, the hypothesis is sufficient and the conclusion is necessary.
3. **Forming the contrapositive by negating only.** That gives the inverse. You must negate *and* swap.

Part II: Practice Problems

Section A — Reading the arrow

1. Rewrite each as an "if. . . then" sentence.

(a) Having a valid ticket is necessary to board the train.

(b) Finishing the project early is sufficient to earn the bonus.

(c) The alarm sounds only if the door is opened.

(d) I will review your draft if you send it by Thursday.

Approved solution

(a) If you board the train, then you have a valid ticket.
(b) If you finish the project early, then you earn the bonus.
(c) If the alarm sounds, then the door was opened.
(d) If you send the draft by Thursday, then I will review it.

(a) and (c) are the two that reverse. "Necessary" names the *conclusion*, and "only if" introduces the *conclusion* — so in both, the clause you read second becomes the "then" part while the arrow still starts from the other clause.
In (d) the "if" clause is written last but is still the hypothesis. Word order in English carries no information about direction; only the connective does.

2. Express the implication "If $ab = 0$, then $a = 0$ or $b = 0$" in each of these ways: (a) using "only if"; (b) using "sufficient"; (c) using "necessary."

Approved solution

(a) $ab = 0$ only if $a = 0$ or $b = 0$.
(b) $ab = 0$ is sufficient for $a = 0$ or $b = 0$.
(c) $a = 0$ or $b = 0$ is necessary for $ab = 0$.
Note that (a) and (b) keep the hypothesis first, while (c) puts the conclusion first. The statement is unchanged in all three.

Section B — The four forms

3. For the implication *"If a quadrilateral is a square, then it is a rectangle,"* write the converse, the inverse, and the contrapositive, and state whether each is true.

Approved solution

Form	Statement	True?
Implication	If a quadrilateral is a square, then it is a rectangle.	Yes
Converse	If a quadrilateral is a rectangle, then it is a square.	No
Inverse	If a quadrilateral is not a square, then it is not a rectangle.	No
Contrapositive	If a quadrilateral is not a rectangle, then it is not a square.	Yes

A single counterexample — a 2×3 rectangle — refutes both the converse and the inverse. That is not a coincidence: those two are logically equivalent, so whatever refutes one refutes the other.

4. For an integer n, consider: *"If n is divisible by* 10*, then n is even."*

(a) Write the contrapositive.

(b) Write the converse.

(c) One of (a) and (b) is true and the other is false. Say which, and give a counterexample for the false one.

Approved solution

(a) If n is not even, then n is not divisible by 10. **True** — it is equivalent to the original, which is true.
(b) If n is even, then n is divisible by 10. **False.**
(c) The converse is false; $n = 4$ is even and not divisible by 10. Any even number that is not a multiple of ten works.

Part (a) required no separate reasoning at all. Once you know the original is true, the contrapositive is true by Theorem 1.46 — and saying so is a complete justification.

Section C — Where it bites

5. A diagnostic test satisfies: *"If a patient has the disease, then the test is positive."*

(a) Write the contrapositive, and say in plain words what it tells a patient.

(b) Write the converse.

(c) A patient tests positive. Does the stated property let you conclude the patient has the disease? Explain.

Approved solution

(a) If the test is negative, then the patient does not have the disease. In plain words: *a negative result rules the disease out.* This is genuinely guaranteed, because the contrapositive is equivalent to the stated property.
(b) If the test is positive, then the patient has the disease.
(c) No. That is the converse, and the converse does not follow from the original. A test that returns positive for everyone would satisfy the stated property perfectly and tell you nothing.

This is the most useful thing in Section 1.3. The stated property makes a negative result

informative and a positive result not. Treating a positive result as conclusive is exactly the converse error, and it is committed constantly outside mathematics.

6. A codebase has the documented invariant: *"If a connection is open, then it appears in the active-connections table."*

(a) A connection does not appear in the table. What may you conclude?

(b) A connection appears in the table. What may you conclude?

Approved solution

(a) That the connection is **not open** — this is the contrapositive, which is guaranteed by the invariant.
(b) Nothing. The invariant permits closed connections to appear in the table — perhaps as stale entries awaiting cleanup. Concluding the connection is open is the converse error.

An invariant of this shape is useful in exactly one direction. Reading it in the other direction is how stale entries turn into bugs.

Section D — Writing

7. In five to eight complete sentences, explain why proving the contrapositive of a statement counts as proving the statement itself, while proving the converse does not. Use the truth table from class, and give one concrete pair of statements to illustrate.

Approved solution

Model answer. An implication and its contrapositive have identical columns in the truth table: in all four rows they take the same truth value, so they are logically equivalent. Because they are the same statement written two ways, establishing either one establishes the other, and a proof of the contrapositive is a proof of the original. The converse is different — it agrees with the original in only two of the four rows, so knowing one is true tells you nothing about the other. Consider "if a number is divisible by 4, then it is even." Its contrapositive, "if a number is not even, then it is not divisible by 4," is also true, as it must be. Its converse, "if a number is even, then it is divisible by 4," is false, as 6 shows. So the contrapositive is a legitimate route to the same destination, while the converse is a different claim that happens to be written with the same two pieces.
Point split guidance. Equivalence of implication and contrapositive, grounded in the table: 3. Non-equivalence of the converse: 2. Concrete illustrating pair: 3.
Weak answer to watch for: "the contrapositive is just the statement backwards." It is backwards *and* negated, and the negation is what makes it work.

Section E — Challenge (optional)

8. Show that the converse and the inverse of $P \Rightarrow Q$ are logically equivalent to each other, *without* building a truth table. Use Theorem 1.46.

Approved solution

The inverse is $(\sim P) \Rightarrow (\sim Q)$. Apply Theorem 1.46 to it, taking $\sim P$ as the hypothesis and $\sim Q$ as

the conclusion. Its contrapositive is obtained by swapping and negating:

$$(\sim P) \Rightarrow (\sim Q) \;\equiv\; \big(\sim(\sim Q)\big) \Rightarrow \big(\sim(\sim P)\big) \;\equiv\; Q \Rightarrow P$$

using double negation (Thm 1.28) at the last step. And $Q \Rightarrow P$ is precisely the converse. So converse and inverse are equivalent.

The neat part: the inverse is the contrapositive of the converse. Once you see that, the pairing in the four-way table stops being something to memorise — there are really just two statements here, each written two ways.

Instructor notes

- **Problems 5 and 6 are the pair that justifies the section.** The diagnostic test and the connection invariant are the same logical error in two settings, and both are errors students will actually commit. Ten minutes on these is worth more than another truth table.
- **Problem 1(a) and 1(c)** are the reversal items. If students get the other two right and these wrong, the issue is English parsing, not logic — give them the drill in Part I rather than more tables.
- **Problem 9** is the elegant one: the inverse *is* the contrapositive of the converse, which collapses four statements into two. Worth putting on the board even for students who skip the challenge.
- **Problem 4(a)** should be answered by citing Theorem 1.46, not by re-deriving. Students who rebuild a table there have not yet internalised that equivalence is transferable.
- Reserved for Homework 3: Section 1.3 Exercises 2, 4, 6, 8, 14 and Section 1.4 Exercises 2, 4, 6, 8 — all even, so no printed answers.